[Gene – protein relationship]
One gene – one enzyme & One gene – one polypeptide concept

Microbiology VI

Introduction

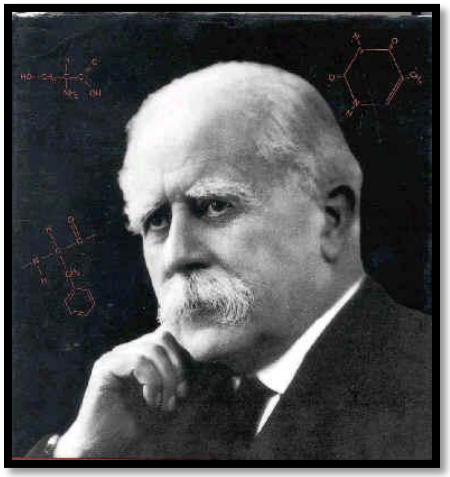
- Today we know that genes encode proteins and protein activity determines **phenotype**.
- However, in the first half of this century, the relationship between genes and proteins was not clear.

- **Mendel** showed that heritable factors determine phenotypes, but he did not know the basis for this relationship.

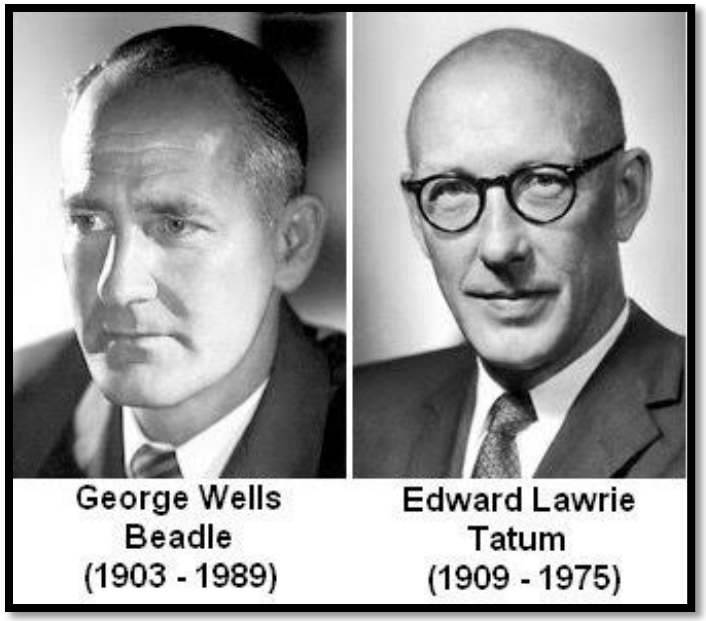
- A series of significant experiments clarified the matter and led to the *central dogma.*
- The central dogma states that genes (DNA) are copied into mRNA, which is used to make protein.

One gene – one enzyme concept

- Two discoveries led to the understanding that genes and enzymes are related in a specific manner.
  1. Archibald Garrod: study of Alkaptonuria – genetically inherited metabolic defect in humans.
  2. Beadle & Tatum: study of Arginine biosynthesis – mutant *Neurospora crassa* (fungi).

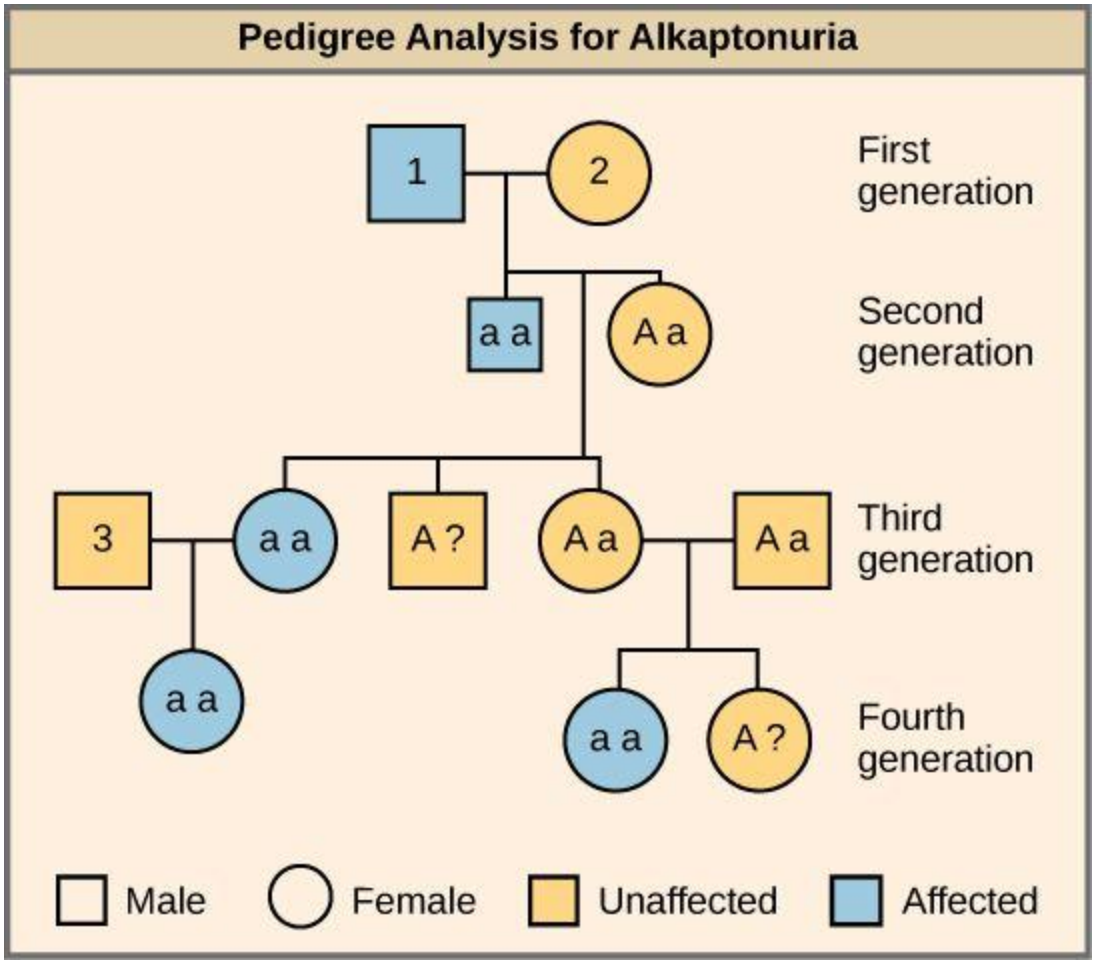


Archibald Garrod



Garrod's Study:

- In 1902, Archibald Garrod an English physician described the genetic inheritance of a metabolic defect.
- The disease he observed was **alkaptonuria**, which causes the urine of affected individuals to turn **black** when exposed to air.
- It Due to defect in enzyme that breaks down the amino acid **tyrosine**.
- Garrod recognized recessive inheritance of the trait in these families; the disease clustered in a few families and was especially likely among the children of first cousins.
- He also determined the identity of the unusual chemical in the alkaptonurics' urine: **homogentisic acid**.


Pedigree Analysis for Alkaptonuria
1
2
First generation
a a
A a
Second generation
3
a a
A ?
A a
A a
Third generation
a a
a a
A ?
Fourth generation
Male
Female
Unaffected
Affected

Phenylalanine

↓

Tyrosine

↓

4-Hydroxy phenyl Pyruvic acid

↓

Homogentisic acid

Homogentisate 1,2-dioxygenase ✕ **ALKAPTONURIA**

↓

Maleyl acetoacetic acid

↓

Fumaryl acetoacetic acid

↓

Fumeric acid + Acetoacetic acid

Black urine (defect)



- Garrod inferred that normal individuals metabolize this chemical into another substance and that alkaptonurics cannot because they lack a particular **enzyme** that is required for metabolism (**degradation**) of homogentisic acid.

- Garrod's study of alkaptonuria and three other diseases contributed to his theory that genetic defects can disrupt metabolic pathways.
- He called these diseases **inborn errors of metabolism.**

BEADLE AND TATUM: ONE GENE–ONE ENZYME

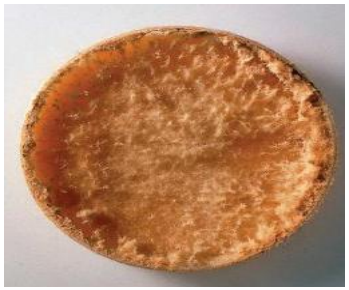
- 1930's - Beadle & Ephrussi studied mutations affecting eye color in **Drosophila.**
- They concluded that each mutation blocks pigment synthesis at a specific step by preventing production of the enzyme that catalyzes that step.
- Specific pathways were not known, so results were inconclusive.
- Their results motivated Beadle to search for the ideal organism to use in extending this work.



Drosophila

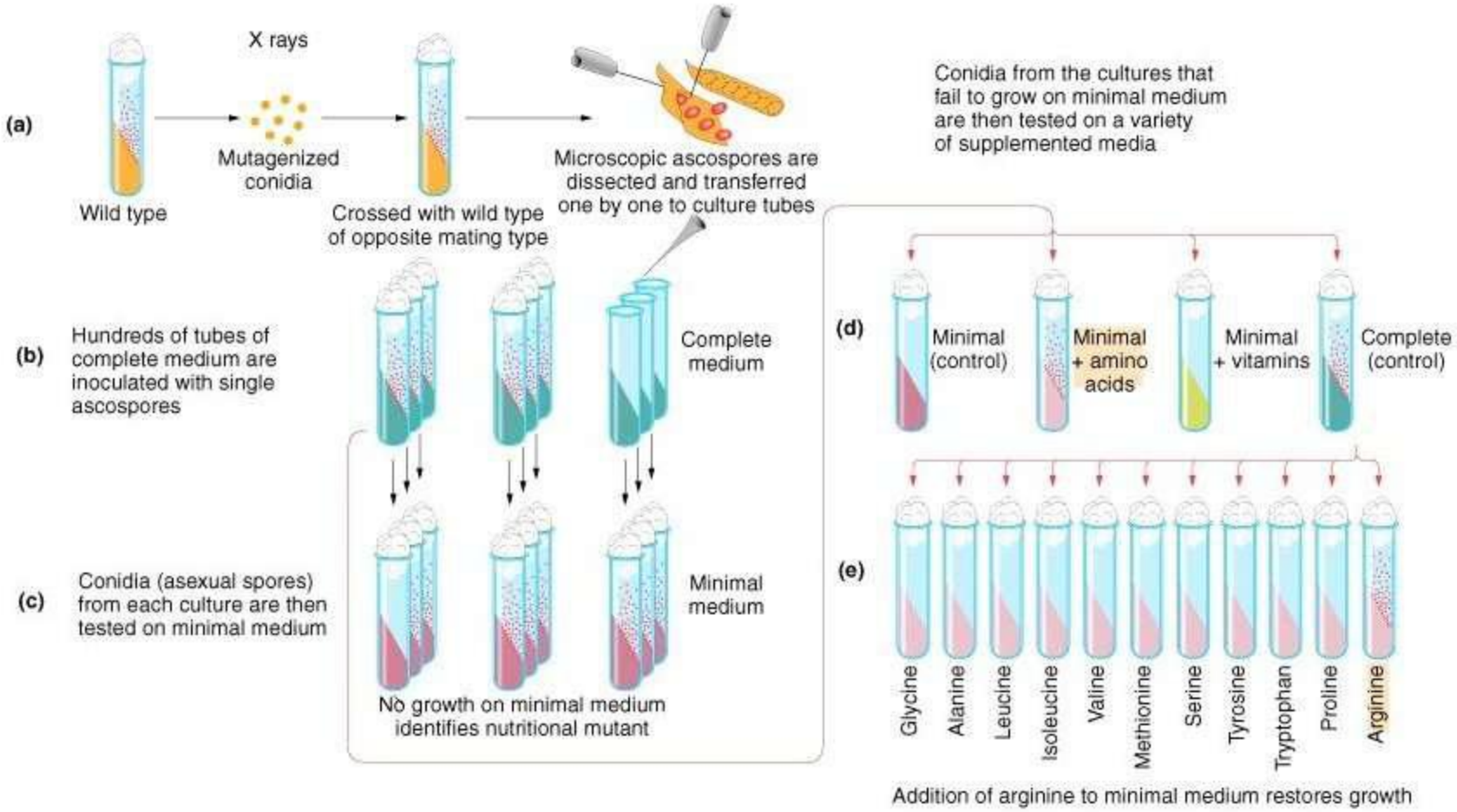
Beadle & Tatum's experiment:

- Treated *Neurospora crassa* (a mold) with X-rays
- Looked for mutations in nutritional requirements:
  - Wild type Neurospora grows on minimal medium (agar enriched with a few nutrients – inorganic salts, sugars & biotin)
  - All mutants will grow on complete medium (agar plus all 20 amino acids & other nutrients)
- Identified the specific amino acid required for growth by each mutant:
  - That identified the defective synthetic pathway
  - Looked at each intermediate step in the blocked synthetic pathway
- Concluded that mutation in a **single gene** blocked production of a **single enzyme**.

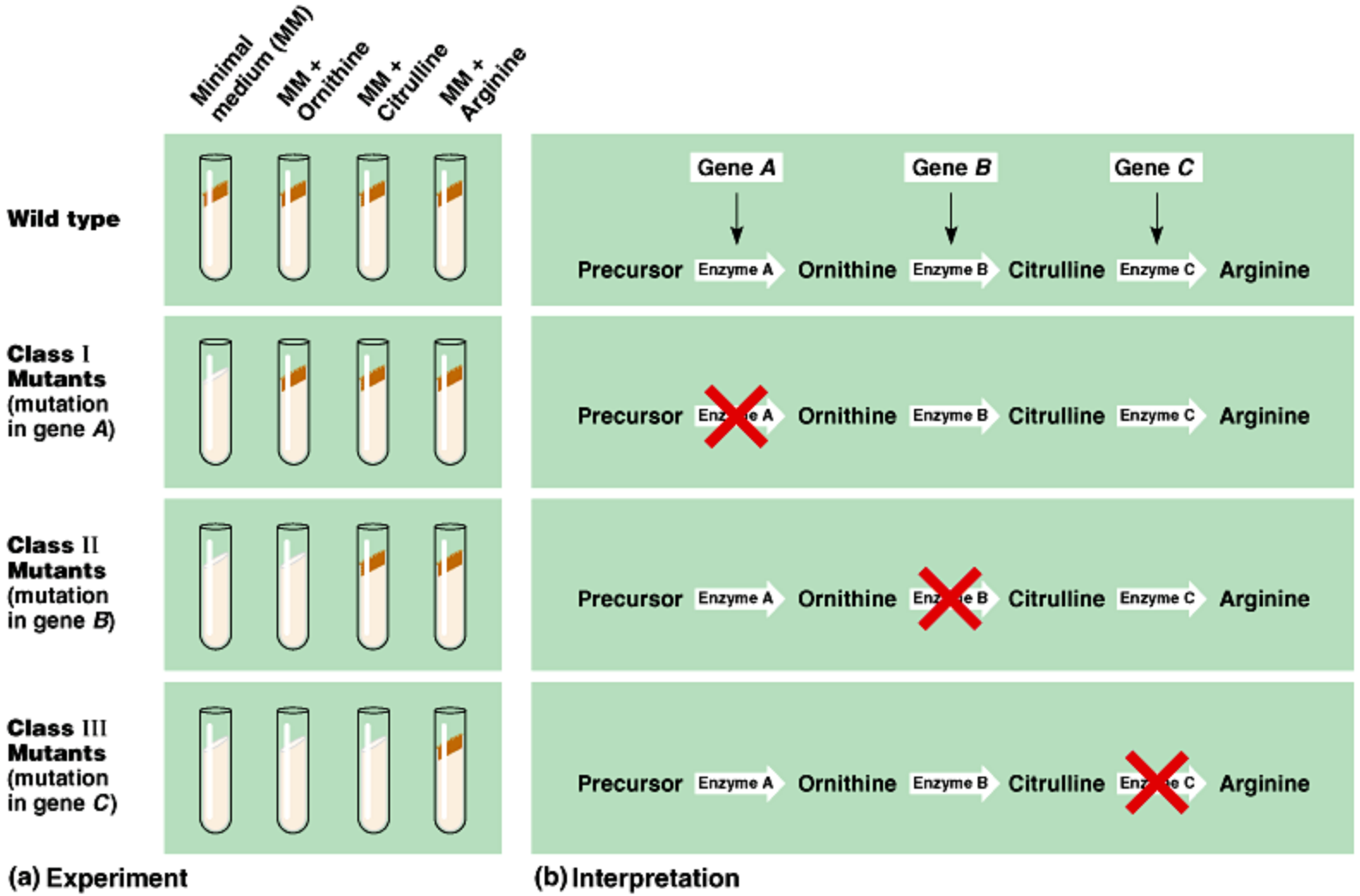


Neurospora

Beadle & Tatum's experiment with *Neurospora crassa*



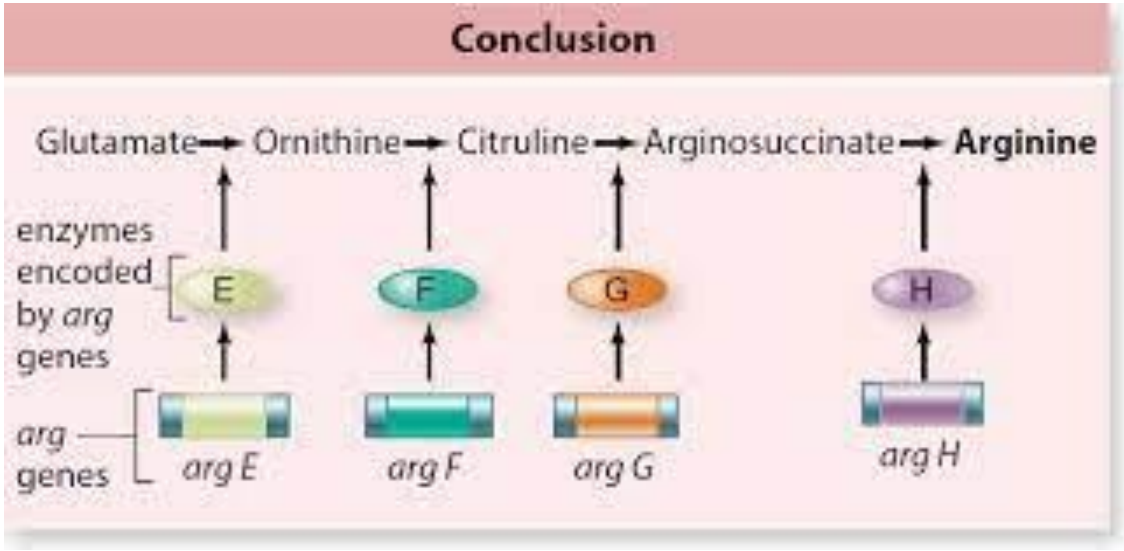
Mutations in genes involved in Arginine biosynthetic pathway



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- Beadle and Tatum tested their strains for their abilities to grow on several compounds related to **arginine**:
- One strain, ***arg-l*** could grow on minimal media that had been supplemented with arginine, ornithine, citrulline.
- A second strain, ***arg-2,*** could grow on minimal media that had been supplemented with arginine or citrulline but not with ornithine.
- A third strain, ***arg-3,*** could only grow in the presence of arginine.
- Citrulline and ornithine were not capable of rescuing the growth defect of the *arg-3 strain.*

- These phenotypes suggested to Beadle and Tatum both a metabolic pathway for these compounds and a role for each gene in that pathway.
- Their explanation was the **one gene-one enzyme hypothesis**.
- It states that each gene controls the activity of a specific enzyme.
- Each of Beadle and Tatum's *arg* mutations identifies a **gene** that **encodes** one of the **enzymes** required for arginine biosynthesis.
- Each **enzyme** converts a **metabolic precursor** into another component in the biosynthetic pathway.


Conclusion
Glutamate
Ornithine
Citruline
Arginosuccinate
Arginine
enzymes encoded by arg genes
E
F
G
H
arg genes
arg E
arg F
arg G
arg H

One Gene – One Polypeptide

- Not all proteins are enzymes.
- Can extend one gene - one enzyme concept to one gene - one polypeptide
- Many proteins are comprised of two or more polypeptides.

- Subsequent to the work of Beadle and Tatum, many enzymes and structural proteins were shown to be heteromultimeric.

- **Heteromultimeric :** It contains two or more different polypeptide chains, with each polypeptide encoded by a separate gene.

For example:

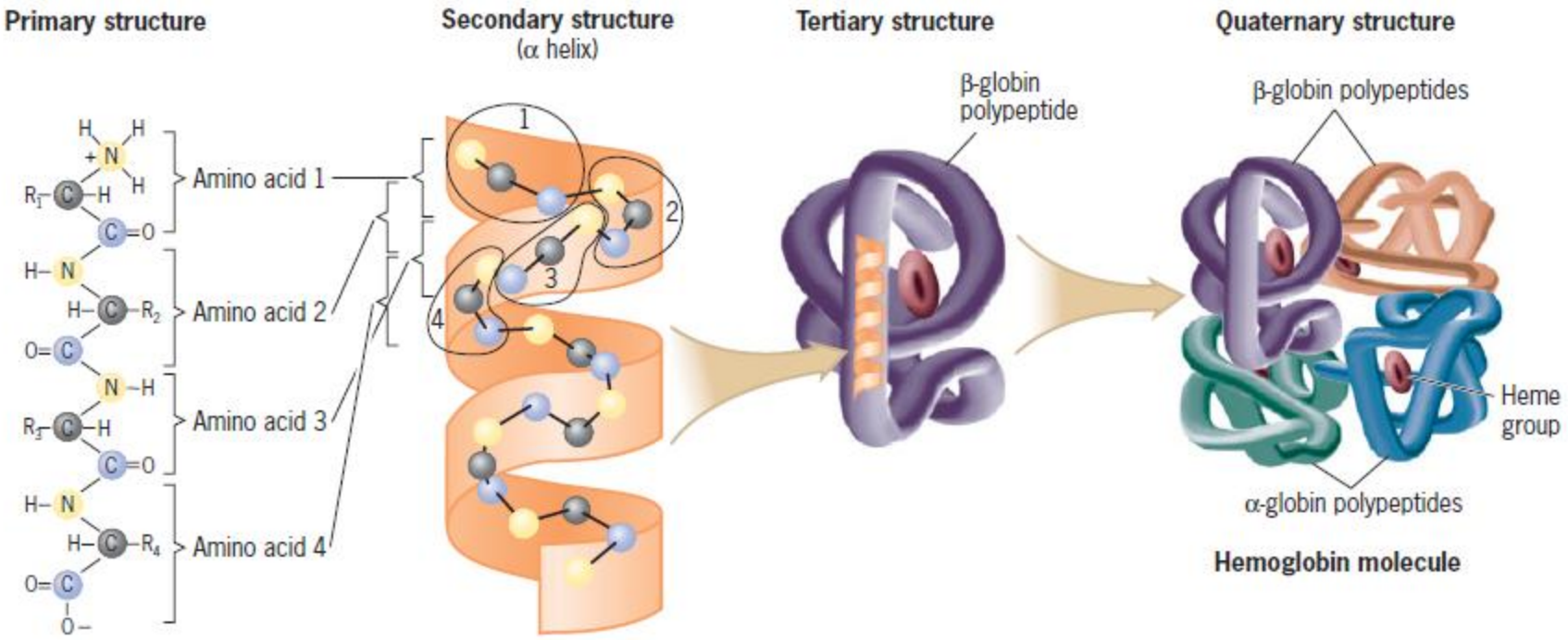
1. Tryptophan synthetase :

- In *E. coli*, the enzyme **tryptophan synthetase** is a heterotetramer composed of two polypeptides encoded by the *trpA gene* and two polypeptides encoded by the *trpB* gene.

2. Hemoglobins:

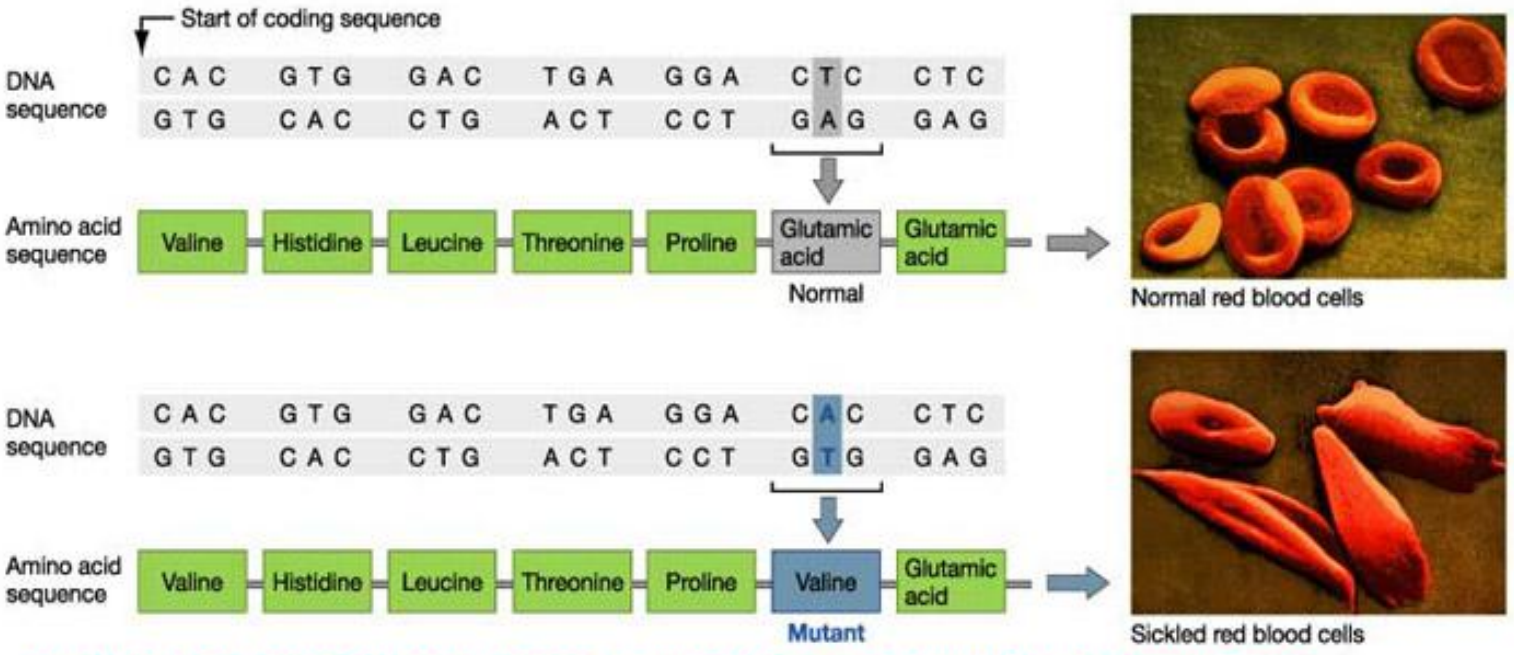
- Which transport oxygen from our lungs to all other tissues of our bodies
- They are tetrameric proteins that contain two α-globin chains and two β-globin chains, as well as four oxygen-binding heme groups

Structure of Hemoglobin



■ **FIGURE 12.3** The four levels of organization in proteins—(1) primary, (2) secondary, (3) tertiary, and (4) quaternary structures—are illustrated using human hemoglobin as an example.

Sickle cell anemia used to study one-gene one polypeptide concept:
Sickle cell RBCs were found to have defect in the amino acid sequence of the hemoglobin molecule.
The mutation of the gene at the **6th** position resulted in **valine** (amino acid) instead of glutamic acid in the normal protein.
As some proteins are made of polypeptides, the one-gene one-enzyme concept was modified to one-gene one-polypeptide concept.



Co linearity of genes & proteins

- The **nucleotide pair** sequences in genes are colinear with the **amino acid** sequences of the polypeptides that they encode.

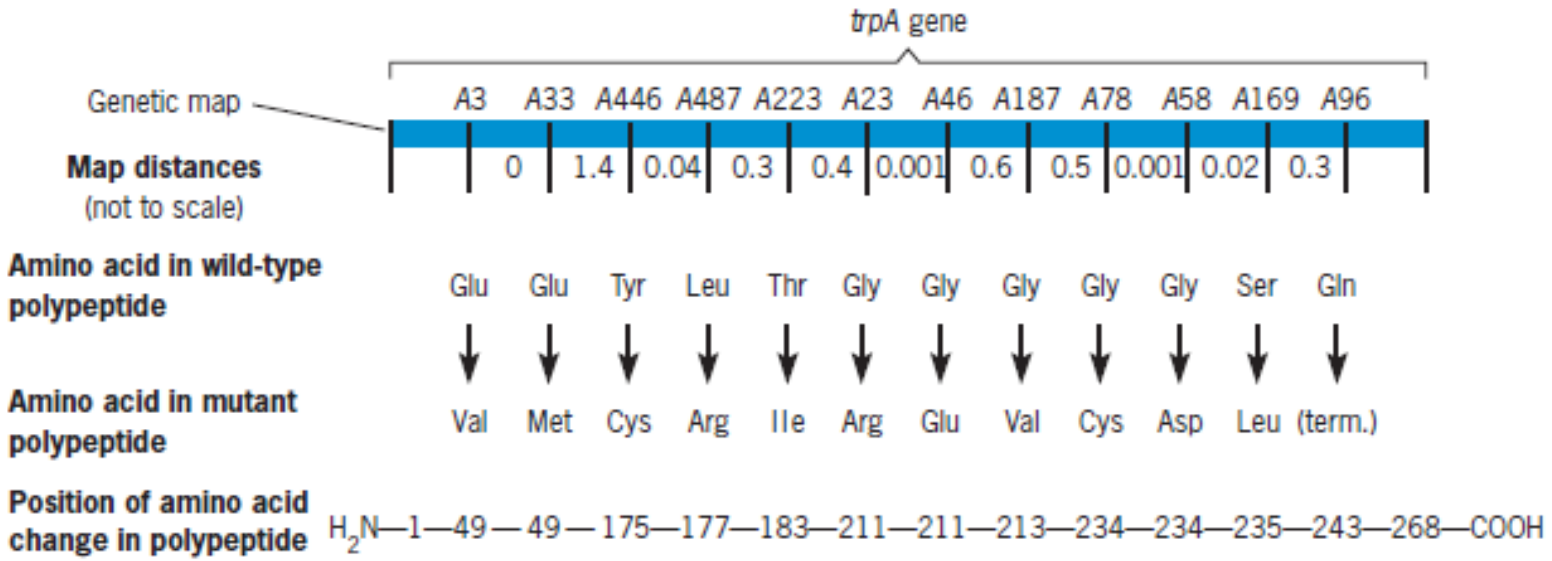
- **Charles Yanofsky** and colleagues suggested the colinearity between a gene and its polypeptide product.

- They studied the on the *E. coli* gene that encodes the subunit of the enzyme **tryptophan synthetase.**

- It contains two polypeptides encoded by the ***trpA gene*** and two polypeptides encoded by the ***trpB gene****.*

- Yanofsky and coworkers performed a detailed genetic analysis of mutations in the ***trpA* gene** and correlated the genetic data with biochemical data on the sequences of the wild-type and mutant tryptophan synthetase polypeptides.

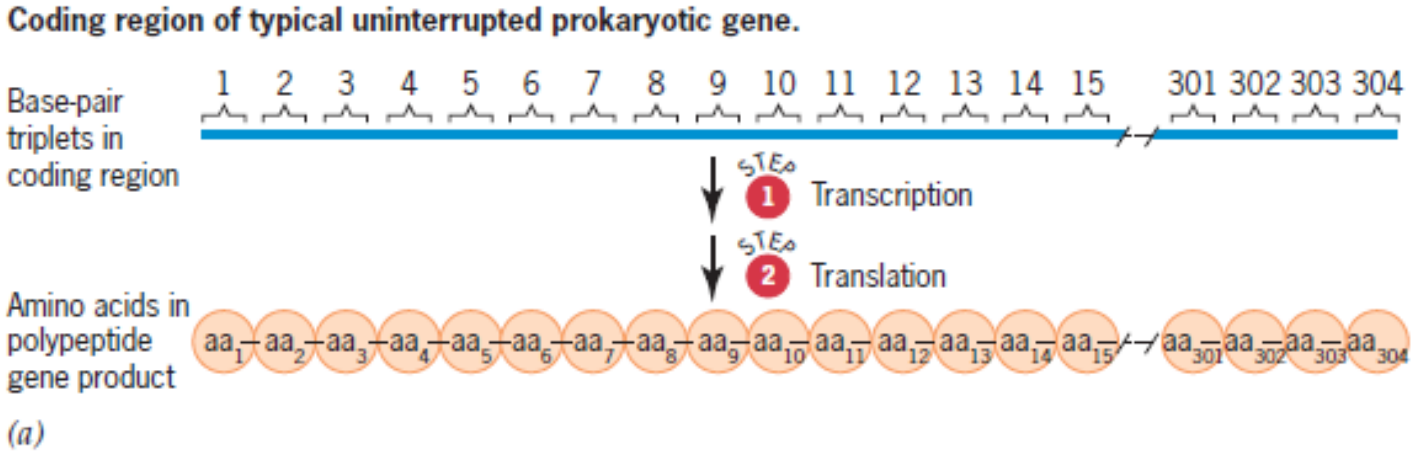
- They demonstrated that there was a direct correlation between the **map positions** of mutations in the ***trpA* gene** and the positions of the resultant **amino acid** substitutions in the tryptophan synthetase **α-polypeptide .**

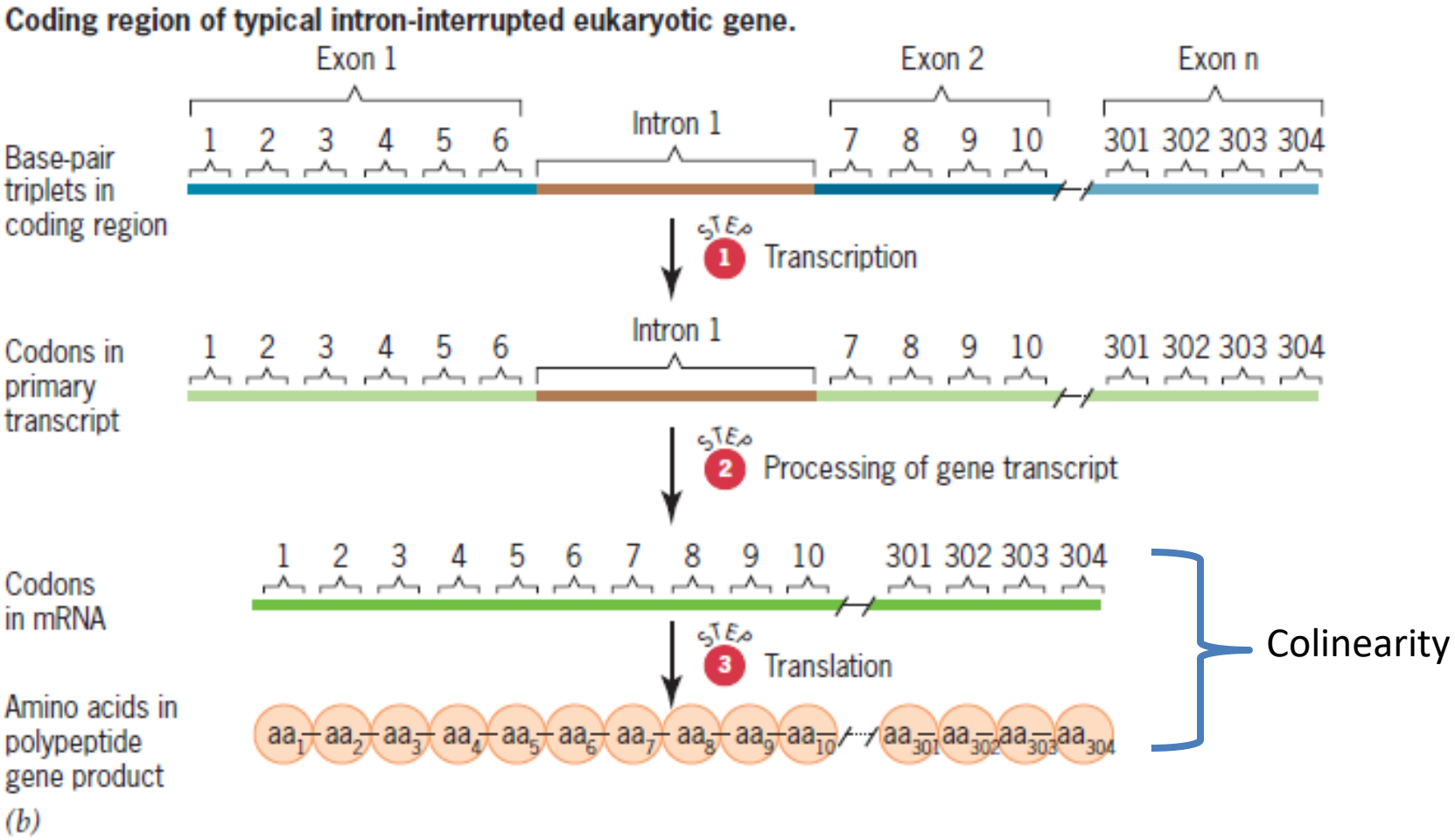


Colinearity between the *E. coli trpA gene* and its polypeptide product, the polypeptide of tryptophan synthetase.

The map positions of mutations in the *trpA* gene are shown at the top, and the locations of the amino acid substitutions produced by these mutations are shown below the map.

Colinearity between the coding regions of genes and their polypeptide products.





■ **FIGURE 12.6** Colinearity between the coding regions of genes and their polypeptide products.